A Full-ranked gene-set enrichment

	Animal models		Human arterial tissue		
	Uraemic calcification	AAD lesions	AD aorta GSE153434	Uraemia / site GSE38750	Uraemia / same site GSE38751
Extracellular matrix organization	**+1.51** *	**+1.65** *	**-1.56** *	**-1.48** *	-1.20
Ossification	+1.34	+1.22	**-1.41** *	**-1.41** *	**-1.26** *
Smooth muscle contraction	+1.42	+0.95	**-2.01** *	-0.90	**-1.60** *
Smooth muscle cell proliferation	+1.19	+1.23	-1.31	-1.03	**-1.58** *
Regulation of smooth muscle cell proliferation	+1.25	+1.28	-1.33	-1.02	**-1.54** *
Positive regulation of smooth muscle cell proliferation	+1.44	**+1.64** *	+1.17	+0.81	-1.23

Orange: positive NES Blue: negative NES * BH q < 0.05 within the original analysis

Signed values preserve direction in grayscale. Animal and human correction families differ.

B Animal ORA (BH q)

	Calcification	AAD
Matrix organization	1.17e-05	4.25e-08
Ossification	9.42e-04	4.10e-05
SM contraction	1.62e-03	8.75e-07
SM proliferation	3.84e-03	2.33e-04
Regulation of SM prolif.	4.49e-03	1.91e-04
Positive regulation	8.57e-03	2.64e-03

GO smooth-muscle contraction is distinct from the KEGG pathway.

C Human sampling design

AD: ascending aorta, 10 vs 10
Uraemia: iliac vs renal artery, 15 vs 19
Uraemia: internal mammary artery, 5 vs 5

GSE38750: disease and site confounded.
GSE38751: same study, not independent replication.
Animal ORA and GSEA are distinct analyses.